Control system MCS
Troubleshooting manual 6410R0360_EN02

ASSA ABLOY

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These operating instructions are an integral part of the door. The original version of these instructions is in German.

Manufacturer
ASSA ABLOY Entrance Systems AB
Lodjursgatan 10
SE-261 44 Landskrona
Sweden
Internet: www.assaabloyentrance.com

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1 Introduction

This troubleshooting guide is provided for troubleshooting the relay controller MCS-R and the frequency converter control MCS-FC with a high speed. They are part of the product and describe its intended and safe use throughout the whole product life cycle.

Before starting troubleshooting on the MCS, read additionally the installation and operating instructions carefully and familiarize yourself with their contents.

The manufacturer is not liable for any damage caused by failure to observe the operating and safety instructions or by unintended use of the MCS.

1.1 Applicable documents

These operating instructions are only valid in combination with the operating instructions for Albany high-speed doors from ASSA ABLOY Entrance Systems and the inspection book. Both documents are supplied with the product.

1.2 Amendments and validity

The information contained in these instructions constitutes the technical specifications applicable at time of publication. Significant changes will be incorporated in a new edition of the troubleshooting instructions. The document and version number of these instructions can be found in the footer.

1.3 Target group

These trouble shooting instructions are addressed to trained, qualified and authorized personnel. All setup configurations, and in particular repairs and maintenance, may only be performed by employees of the manufacturer or other trained specialists.

2 Safety instructions

Before installation, operation, service, or maintenance work on this high speed door or their electrical components it is mandatory to read and to comply with the safety instructions.

In addition to the general safety instructions in this section, always observe the safety instructions related to specific actions that are included in other sections. In particular, the general safety instructions in the operating instructions for the door are also applicable.

A nonobservance of these safety instructions can lead to damage of the door, injury or death.

2.1 General safety instructions

- Observe the general, legal and other nationally valid regulations for accident prevention and the accepted technical rules for safety-oriented work.
- All work on electrical systems and connections may be performed only by qualified personnel with appropriate authorization.
- The electrical work must be carried out according to the applicable local standards and legal regulations.
- Before performing any work on the control unit, cables, connectors or electrically live components, always disconnect and secure the CEE plug or turn of main switch.
- Make no change to the control unit or the safety devices which may affect the safety of the door.
- In order to avoid damage by electronic discharge, do not touch the circuit board of the control system, the display and the auxiliary modules without appropriate ESD protective equipment.

2.2 Safety instructions for installation and maintenance

- Ensure that neither persons nor vehicles pass through the door or are in the door hazard area during installation and maintenance.
- Dangerous contact voltages are present at the frequency converter up to 180 seconds after switching off/disconnecting the supply voltage (discharge of the capacitors). You must therefore wait at least 180 seconds after disconnecting the supply voltage before working on electrical parts, wiring or opening the frequency converter housing.
  Non-observance of this rule can lead to electric shock, serious injury or death.
- You should generally avoid touching electronic components on the MCS board or auxiliary modules.
- When opening the housing cover of the MCS or the frequency converter, ensure that the control system is protected against moisture.

2.3 Safety instructions for “hold to run mode”

- In hold to run mode, all safety devices are disabled. According to the operating mode it results in the following door behavior:

  Operating mode AUTOMATIC ↑/↓:

  The hold to run mode is activated only when the safeties are activated. When closing the door, the door moves as long as the close button is activated and the bottom end position is not reached. The opening movement is done by pressing the open button in automatic mode. Upon reaching the upper or lower end position the door movement is stopped.

  Operating mode AUTO↑/DEADMAN↓:

  When closing the door, the door moves as long as the close button is activated and the bottom end position is not reached. The opening movement is done by pressing the open button in automatic mode. Upon reaching the upper or lower end position the door movement is stopped.

  Operating mode DEADMAN ↑/↓:

  The door moves during the opening or closing movement only as long as the open or close button is activated and the end positions are not reached.
- Operate the door only if door area is visible.
- Do not operate the door is any person or object is located in the hazard area.

3 Read out event log and software version

3.1 Read out of the event log

The event log stores the error messages. At total 128 messages can be stored. If the 128th message is reached the 64 oldest entries will be deleted. Date and time of the error and the plain text message will be shown.

Display of the error log:

LANGUAGE => SERVICE => SERVICE EVENT HISTORY

3.2 Clear event log



The event log should be cleared only by ASSA ABLOY Entrance systems, because all previous error message will be permanently deleted.

Clear event log:

LANGUAGE => SERVICE => SERVICE CLEAR EVENT HIST

3.3 Read out software version

In the control system display the used software version can be read out.

LANGUAGE => SERVICE => SERVICE SOFTWARE VERSION

For the MCS-FC the top display row shows the software version of the control system and in the bottom line the one from the frequency converter.

4 Reset of the control to factory setting

To reset all parameters to factory setting successively the following two steps must be performed:

1.) LANGUAGE => SERVICE => SERVICE CFG SOFT RESET

2.) LANGUAGE => SERVICE => SERVICE FACTORY RESET

After confirming the two previous steps the control system must turn off for min. 10 seconds by pulling the CEE-plug, so that all settings are taken (discharge time of capacitors in the FC). The event log is not erased by this step.

5 Status indicator

For many messages the first display row shows the current status, e.g. BLOCKED. This is marked in the manual with (Status)*.

Below are all status messages of the control and their meanings.

BLOCKED	Control system blocked and no door movement possible, because operating mode deactivated by error.
STEP MODE	Active, if the door position is outside of the stored position range. The door can be driven only in the direction which returns it back to the valid position range.
DEADMAN DOWN	For closing movement only hold to run mode possible. This message appears when there are problems with the safeties (wrong / missing programming, activated safeties ...).
INV UPDATE	Message appears during automatic parametrization of the frequency converter by the control system.
PULL-IN PROTECT	Message is not used at high-speed door of the Albany brand. Check the setting of the safeties.
AUTOMATIC UP/DWN	Normal operating display with automatic operation in both directions.
DEADMAN UP/DOWN	Hold to run mode active in both directions.
TWO WAY TRAFFIC	Message appears when operating with direction detection traffic lights.
OPERATION MODE	Initialization of the control system. Control system checks which parameters need to be send to the frequency converter.

6 Errors without display message

6.1 Door does not close / Hold open time does not start

Error	Correctives
Permanent open command on the control system. The hold open time does not expire.	Check all connected actuators.
The automatic closing function is deactivated and the control is waiting for manual closing signal.	Activate operating mode auto closing. OPERATING MODE => select function and adjust hold open time to a value greater 0 seconds. SPECIAL SETUP => AUTO CLOSE ENABLED => CLOSE DELAY XX (Setting value for hold open time).
The selected operating mode is AUTO ↑ / DEADMAN ↓ or DEADMAN ↑/↓	Change operating mode to AUTOMATIK ↑/↓.
Not used safety inputs are not bridged. The hold open time in the display doesn't start to run or the hold open time is not shown „-----„	Insert jumper on KL23 3/4 and 4/5.
The safeties have been disabled in the software during the setup.	Adjust parameter for KL1, KL4 and KL5 according default in the basic wiring diagram.
The safety edge has been activated with more actuation then set in the software as max. value. Door lock in the top position and hold open time is not shown „-----„.	See display message „REVERSED 3 TIMES“ (Chapter 7.28)

6.2 Door open and close delayed

Error	Correctives
A lead/pre-warning time before opening and closing has been adjusted.	Deactivation of the lead time. SPECIAL SETUP => SPECIAL SETUP TRAFFIC LIGHT => TRAFFIC LIGHT or WARNING LIGHTS or TRAFFIC LIGHT TWO WAY TRAFFIC => Set lead time to 0s. or SPECIAL SETUP => SPECIAL SETUP RELAYS => RELAYS 1-3 => R1-3 TYPE

	MOVEMENT => R1-3 PRERUN => Set prerunning time to 0s or SPECIAL SETUP => SPECIAL SETUP RELAYS => RELAYS 1-3 => R1-3 TYP AUTO CLOSE => R1-3 PRERUN => Set prerunning time to 0s.

6.3 Door close only in hold to run mode

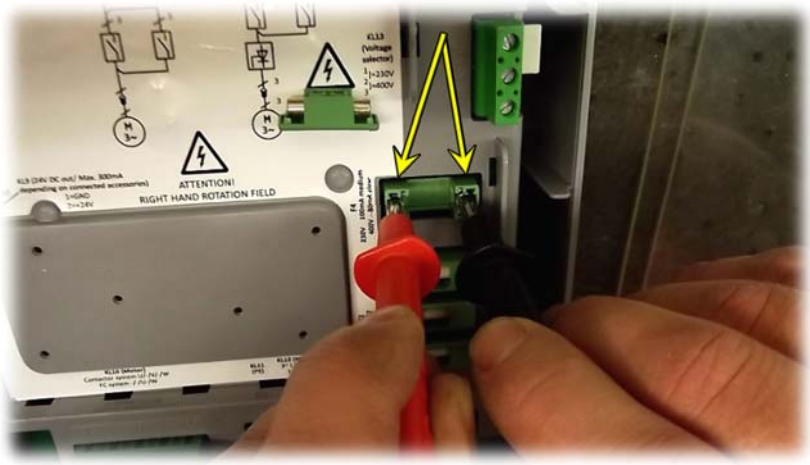
Error	Correctives
The operating mode of the door has been programmed before the settings of the safeties or the safeties were not or not complete programmed.	Parametrization of the safeties. SAFETY DEVICES => Adjustment of all parameters according default values in basic wiring diagram.
The operating mode DEADMAN ↑/↓ or AUTO ↑/DEADMAN ↓ was adjusted.	Change to operating mode AUTOMATIC ↑/↓ during setup.

6.4 Fuse F4 is blown

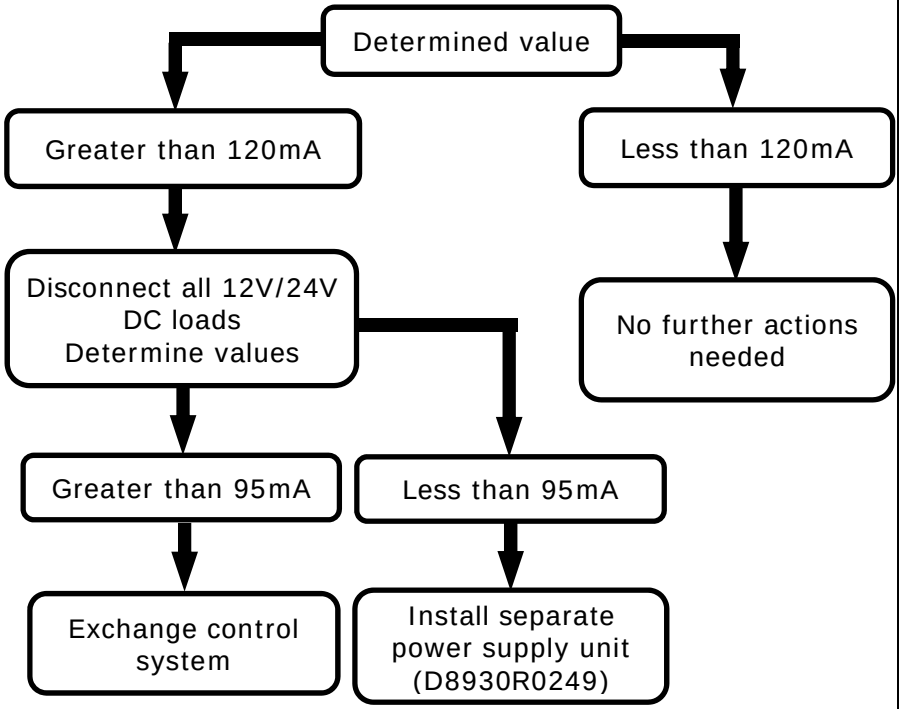
Error	Correctives
The fuse F4 in the control system is blown.	See Chapter 6.5 from point 2

6.5 Dark display <u>without</u> any text

Error	Correctives
The display is complete dark and without any text display.	1.) Check net voltage on KL12. 2.) >Check voltage selector mounting on KL13. 230V => 1/2, 400V => 2/3 3.) Check fuse F4 in the control system (MCS-R=80mA slow, MCS-FC=100mA medium). 4.) Short circuit in 24V circuit. Check all 24V components and cabling for short circuit. 5.) The flat cable from display is not connected to the display plug in the control system. **The following points are only for the MCS-FC control.** 6.) Measure current consumption of the control system => Disconnect CEE plug, remove fuse F4, Mount jumper from voltage selector to KL13 1/2. Turn on control system, select measurement range of measuring tool to 300/400mA AC (alternating current) and measure current consumption on both contacts in the base of the fuse F4.



Note value and proceed according the following flow chart:



6.6 Blue display without any text

Error	Correctives
The display lights up blue without any text.	1.) In the setting for the frequency converter a wrong parameter has been selected (Select parameter „FS“ instead of „RUE”). MOTOR SETUP => MOTOR SETUP MOTOR CONTROLLER => Selection according default in basic wiring diagram. 2.) The plug from display cable was connected in the wrong direction in the control system. The red wire needs to be on the left side (see basic wiring diagram). 3.) The voltage selector KL13 at the MCS-R has been inserted at 230V instead of 400V. The blue display lights only very weakly without any text indication. MCS-FC (230V) => KL13 1,2 connected MCS-R (400V) => KL13 2,3 connected

6.7 Motor is very hot

Error	Correctives
The motor is very hot during operation or runs against the brake.	1.) The wrong brake function was selected (power-on/power-off brake). Setting the correct brake function on slide switch S1 in the frequency converter according basic wiring diagram. Attention RR300 Retail Power-on brake! (See user manual chapter 4.2.3 Selection power-on or power-off brake). Worn brake needs to be replaced. 2.) The wrong brake voltage was set in the converter on KL2. Attention RR300 Retail 100V brake! Wiring change in converter on KL2 necessary! (KL2:1+3=200V, 2+3=100V). Worn brake needs to be replaced. 3.) Check mechanic of the door. 4.) Only for MCS-R: The motor is supplied only with 2 phases => Measure voltage on KL10 U,V,W => Check fuses F1-F3 if phase is missing => If all fuses ok check

	net voltage on KL12 L1,L2,L3 => If ok please replace control system.

6.8 Not possible to program end positions

Error	Correctives
The end positions cannot programmed during setup of the control system.	1.) The unused inputs KL7 1/2 (Door safety), KL20 3/4 (push button), KL23 3/4/5 (loop detector) need to be bridged. If a sensor is connected to KL7, KL20 or KL23 check sensor circuit. 2.) The safety chain in the motor (KL8) is interrupted. Check safety chain (see chapter 7.16) 3.) The parameter motor controller for the MCS-FC was set to MOTOR CONTROLLER CONTACTOR/RELAY instead MOTOR CONTROLLER INVERTER FS. For MCS-FC change setting to MOTOR CONTROLLER INVERTER FS.

6.9 Encoder value not shown/changing during setup

Fehler	Abhilfe
During setup of the endpositions the door position (encoder value) is not displayed or the value doesn’t change during door movement.	1.) See chapter 7.17

6.10 After programming of the end positions the door does not close

Error	Correctives
After programming of the end positions the door cannot be closed or the hold open time will not start.	1.) The input KL23 3/4/5 (loop detector) need to be bridged if not used. If a sensor is connected to KL23 check sensor circuit.

6.11 Door runs to fast or to slow

Error	Correctives
The door runs during the movement to fast or to slow.	1.) The wrong parameter for the frequency converter has been selected during the setup. Set parameter according default in basic wiring diagram. 2.) The used drive system has got a wrong gearbox ratio. Change drive to right type.

7 Error with display message



For many messages the first display row shows the current status, e.g. BLOCKED. This is marked in the manual with (Status)*.

7.1 Stop

Error message:	(Status)* STOP
Reason:	Stop is activated.
Correctives:	1.) Check stop buttons. 2.) Not used stop inputs should be bridged on KL20 3/4.

7.2 Relay defect

Error message:	(Status)* RELAY DEFECT
Reason:	The test of the internal relay was faulty. The voltage selector (KL13) has been inserted incorrectly. The frequency converter is defect or has a short circuit. In the electrical power grid, there was briefly a short collapse of the main voltage.
Correctives:	1.) Check fuses F1-F3 in the control system and replace if necessary. Check frequency converter board by optical inspection. Exchange if there are black or burnt components. Before turn on power again check setting of voltage selector KL13 (MCS-FC 230V)! 2.) Reset of the error message under SERVICE => SERVICE CLEAR ERRORS and turn off main power for min. 10 seconds. 3.) Replace control system.

7.3 OSE 1 triggered

Error message:	(Status)* OSE 1 TRIGGERED
Reason:	The input OSE 1 (KL3) was activated.
Correctives:	1.) Disable the input OSE 1 (KL3) in the settings, as this is not used in all Albany ASSA ABLOY high speed doors. Disable => SAFETY DEVICES => OSE1 SETUP KL3 => adjust to DISABLED (see also user manual chapter 4.8.8)

7.4 OSE 1 error

Error message:	(Status)* OSE 1 ERROR
Reason:	The input OSE 1 (KL3) has triggered.
Correctives:	1.) Disable the input OSE 1 (KL3) in the settings, as this is not used in all Albany ASSA ABLOY high speed doors. Deactivation SAFETY DEVICES => 4-W PHOTOC. KL5 => OSE1 SETUP KL3 => adjust to DISABLED (see also user manual chapter 4.8.8)

7.5 OSE 2 triggered

Error message:	(Status)* OSE 2 TRIGGERED
Reason:	The safety on input OSE 2 (KL4) has been triggered.
Correctives:	1.) Check whether the light beam of the connected photocell or detection range of radar / infrared detector is blocked by persons or objects. Optionally adjust detector, clean and align photocell. 2.) Check wireless safety edge WDD (only for RR200 door). 3.) Check wiring of the sensor circuit on KL4. 4.) Check settings according default parameters in basic wiring diagram. 5.) Not used input should be bridged between KL4; 2 and 4. 6.) Short circuit in 12 V DC control circuit (KL4, KL5;2 (Attention: KL5;2 test output with 12V, KL5;1 with 24V)). Check connection and cabling on short circuit.

7.6 OSE 2 error

Error message:	(Status)* OSE 2 ERROR
Reason:	The safety device on input OSE 2 (KL4) has failed the safety test.
Correctives:	1.) Cleaning and possibly aligning of the connected safety on KL4. 2.) Check the wiring of the sensor circuit on KL4. For test disconnect the sensor and mount a bridge between Kl4; 2 and 4. If the problem persists => see 6. 3.) Check the settings according to the default value in the basic wiring diagram. 4.) Not used input should be bridged between KL4; 2 and 4. 5.) The receiver of the photocell is not turned off during the test of the safeties, because it is influenced e.g. by the light of another light source (e.g. photocell/light curtain, laser

	scanner etc.). Spatial separation of the optical light sources or foreclosure, thus no interference occurs. 6.) Check control system outputs with measurement device. Measure KL4; 2 (Test) and KL4; 3 (+12V DC) contrary KL4; 1 (GND). If 12V are missing on one of the both terminals replace control system.

7.7 Safe.edge 1 trig.

Error message:	(Status)* SAFE.EDGE 1 TRIG.
Reason:	The safety edge on input KI2; 1 and 2 was activated or disconnected.
Correctives:	1.) Check complete cabling of the safety edge circuit. 2.) If this message appears when the closing door reaches the bottom end position, correct maybe the bottom endposition and the switch-off point for the safety edge. Setting under: ADJUST ENDPOS. => ADJUST ENDPOS. BOTTOM => Adjust bottom endposition ADJUST ENDPOS. => ADJUST ENDPOS. PRE END POSITION. (see also user manual chapter 4.8.3) The exact default value for the pre-end limit switch setting is shown in the basic wiring diagram parameter list. 3.) Measure safety edge resistance. Therefor disconnect safety edge on plug KL2 and measure the resistance between both cable ends. This must be in the range of approximately 8.2 KΩ. For very big resistance values check the circuit for open circuit. If necessary replace the safety edge. 4.) Check safety edge input of the control system. Therefor disconnect the safety edge and connect a 8,2 kΩ resistor between KL2; 1 and 2. If the error persists, replace control system.

7.8 Safe.edge 1 err.

Error message:	(Status)* SAFE.EDGE 1 ERR.
Reason:	The safety edge connected on control system input KL2; 1 and 2 is damaged.
Correctives:	1.) Check complete cabling of the safety edge circuit. 2.) Measure safety edge resistance. Therefor disconnect safety edge on plug KL2 and measure the resistance between both cable ends. This must be in the range of approximately 8.2 KΩ. For strong deviation resistance values check the safety edge circuit. If necessary replace the safety edge. 3.) Check safety edge input of the control system. Therefor disconnect the safety edge and connect a 8,2 kΩ resistor between KL2; 1/2. The error can be reset in the upper endposition by a renewed open pulse. If the error persists, replace control system. 4.) Continuous operation/short circuit during the test of the safety edge in the top endposition. Check safety edge and safety edge circuit on short circuit. The error can be reset in the upper endposition by a renewed open pulse.

7.9 Safe.edge 2 trig

Error message:	(Status)* SAFE.EDGE 2 TRIG
Reason:	Wrong parameter setting during the setup, because this input is not used on Albany ASSA ABLOY high speed doors. The safety edge on input KL2; 3 and 4 was activated or disconnected.
Correctives:	1.) For deactivation disconnect the input in the settings. SAFETY DEVICES => SAFETY 2 KL2 3-4 => adjust to DISABLED.

7.10 Safe.edge 2 err.

Error message:	(Status)* SAFE.EDGE 2 ERR.
Reason:	Wrong parameter setting during the setup, because this input is not used on Albany ASSA ABLOY high speed doors. The safety edge on input KL2; 3 and 4 of the control system is defect.
Correctives:	1.) For deactivation disable it in the settings. SAFETY DEVICES => SAFETY 2 KL2 3-4 => adjust to DISABLED.

7.11 4W-photoc. trigg

Error message:	(Status)* 4W-PHOTOC. TRIGG
Reason:	The photocell/light curtain on input KL5 was activated.
Correctives:	1.) Check whether the light beam of the connected photocell / light curtain is blocked by persons or objects. 2.) Clean and if necessary align photocell / light curtain on KL5. 3.) Check cabling of the photocell / light curtain on KL5. 4.) Check settings according default parameters in basic wiring diagram. Additionally the disconnection point of the doorline photocell should be adjusted to approx. 350mm height. SAFETY DEVICES => 4-W PHOTOC. KL5 => … => DISABLE PHOTOC. 5.) Not used inputs need a wire bridge between KL5; 2 and 3. 6.) Short circuit in 12V control circuit (KL4, KL5;2 (Attention: KL5;2 test output 12V, KL5;1 24V)). Check cabling and connection for short circuit.

7.12 4W-Photoc.error

Error message:	(Status)* 4W-PHOTOC. ERROR
Reason:	The safety test of the photocell / light curtain on KL5 was faulty.
Correctives:	1.) Clean and if necessary align photocell / light curtain on KL5. 2.) If input not used insert a wire bridge between KL5; 2 and 3. 3.) Check cabling of the photocell circuit on KL5. For test disconnect complete photocell / light curtain and insert wire bridge between KL5; 2 and 3. If the problem persists => see 5. 4.) Check all settings according default parameters in basic wiring diagram. 5.) The receiver of the photocell is not turned off during the test of the safeties, because it is influenced e.g. by the light of another light source (e.g. photocell/light curtain, laser scanner

	etc.). Spatial separation of the optical light sources or foreclosure, thus no interference occurs. 6.) Check control system outputs with measurement tool. Measure KL5; 2 (Test 12V DC) and KL5;1 (+24V DC) against KL5;4 (GND). If the 12V or 24 V are missing on one of the two terminals change the control system. Check parameter settings according default values in basic wiring diagram.

7.13 Photocell trigg.

Error message:	(Status)* PHOTOCELL TRIGG.
Reason:	Wrong parameter setting during the setup, because this input is not used on Albany ASSA ABLOY high speed doors. The 2-wire photocell on input KL6 was activated.
Correctives:	For deactivation disable the input in the settings. SAFETY DEVICES => 2-W PHOTOC. KL6 => Set DISABLED (see user manual chapter 4.8.4)

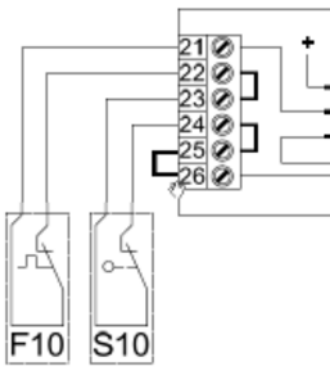
7.14 4W-Photoc. error

Error message:	(Status)* 4W-PHOTOC. ERROR
Reason:	Wrong parameter setting during the setup, because this input is not used on Albany ASSA ABLOY high speed doors. The safety test of the 2-wire photocell on input KL6 was faulty.
Correctives:	For deactivation disable the input in the settings. SAFETY DEVICES => 2-W PHOTOC. KL6 => Set DISABLED (see user manual chapter 4.8.4)

7.15 Door safety

Error message:	(Status)* DOOR SAFETY
Reason:	The emergency stop or crash sensor on KL7 was activated.
Correctives:	1.) Check emergency stop input and cabling on KL7. 2.) Some door types use this input for the crash sensor detection. Thread the curtain and bottom profile according door user manual. The error will be reset automatically after threading of the profile and the door is ready for operation again.

7.16 Motor safety

Error message:	(Status)* MOTOR SAFETY
Reason:	The safety chain in the drive unit is interrupted (KL8).
Correctives:	1.) Check cabling on input KL8. 2.) The thermal contact from motor, handlever/crank and if applicable the hand chain drive are connected in the motor safety chain in serial (terminal 21-26 in the limit switch box of the motor). Not used terminals in the motor limit switch box need to be bridged by wire.  F10= Thermal contact in motor

	S10= manual handlever/crank Terminal 25, 26 not used and bridged with wire.

7.17 Positioner

Error message:	(Status)* POSITIONER
Reason:	The digital limit switch (encoder) or the communication was disturbed/interrupted (KL17), or wrong parameter has been adjusted during the setup.
Correctives:	1.) Adjust parameter for the encoder according default parameters in basic wiring diagram. 2.) The plug from the digital limit switch in the motor was plugged to KL18 instead KL17 => Plug to KL17. 3.) If this message is alternating displayed with the message „(Status)* UNAUTHORIS. MOVE„ the control system was not complete programmed. => Complete setup necessary. 4.) Check cabling and wire connection in the digital limit switch plug if this was disconnected before (e.g. for wall entrance). 5.) Sporadic occurrence of the error message during door movement => add adapter cable (for more details see point 6.). 6.) Check digital limit switch in the motor. Therefor change to endposition adjustment in the setup: ADJUST ENDPOS. => ADJUST ENDPOS. TOP and check displayed value. Door opens => Value must increase Door closes => Value must decrease If this is done in this way the encoder is ok. If no value is displayed or the value will not change during door movement the communication between encoder and control system is interrupted. => Check wiring => Plug adapter cable with bias resistors between encoder cable from motor and KL17 in the control system (only necessary for MCS-FC control systems delivered from November 2016 till approx. March 2017). As of version V3.2 (see MCS nameplate) the adapter cable is no longer required. => Change encoder.

	7.) Short circuit in 12 V control circuit (KL4, KL5;2 (Attention: KL5;2 test output 12V, KL5;1 24V)). Check if short circuit in cabling and connection. 8.) Replace MCS control system.

7.18 Battery empty

Error message:	(Status)* BATTERY EMPTY
Reason:	The storage battery in the digital limit switch (encoder) is discharged. Message only occurs with Dynacompte and Dalmatic encoder model.
Correctives:	1.) Replace digital limit switch in the drive unit. 2.) Limit switch cable broken. Replacement necessary.

7.19 Battery low

Error message:	(Status)* BATTERY LOW
Reason:	The battery voltage of the storage battery in the digital limit switch (encoder) is low. Message only occurs with Dalmatic encoder model.
Correctives:	1.) Replace digital limit switch in the drive unit.

7.20 Inverter error

Error message:	(Status)* INVERTER ERROR
Reason:	The communication between control system and frequency converter is disturbed or interrupted.

Correctives:	1.) The red LED in the FC should flash with applied supply voltage. If its flashes go to point 4.) else point 2.). 2.) Check and if applicable replace fuse F1 and F3 in the control system. 3.) Check 230 V supply voltage on KL12; 1 and 3. 4.) Check communication cable between control (KL18) and frequency converter (KL4). 5.) Short circuit in 12V control circuit (KL4, KL5;2 (Attention: KL5;2 test output 12V, KL5;1 24V)). Check if short circuit in cabling and connection. 6.) The brake resistor in the frequency converter (200 Ω) is broken. To check turn of power supply. **Warning the frequency converter performs up to 180 seconds after turn off a dangerous voltage!** Open the frequency converter housing and disconnect both wires of the brake resistor from the board. Measure the resistance between both wires. If the value is significant greater or less than 200 Ω the resistor is broken and the complete frequency converter needs to be replaced. 7.) The frequency converter is too hot. Reduce number of door cycles. Mount frequency converter outside thermal interferences (solar radiation, heat radiation drive unit …).

7.21 Out of range

Error message:	(Status)* OUT OF RANGE
Reason:	The final position is outside the programmed valid range.
Correctives:	1.) Check and if applicable replace fuse F1 and F3 in the control system. 2.) The rotational direction from digital limit switches (incremental encoder in the motor) is set incorrectly. MOTOR SETUP => MOTOR SETUP ENCODER DIR. => Adjustment according default parameter in basic wiring diagram. 3.) The door has passed the end positions. Move door with step mode back to normal position between upper and lower end position.

	4.) The brake is worn and slips („cannot hold“). Check function of brake and if applicable replace.

7.22 Missing endpos.

Error message:	(Status)* MISSING ENDPOS.
Reason:	The end positions are not stored or don’t have a logic value.
Correctives:	1.) The end positions have not been re-stored after changing the encoder direction. ADJUST ENDPOS. => ADJUST ENDPOS. TOP + ADJUST ENDPOS. BOTTOM **and** store disconnection point of doorline photocell new SAFETY DEVICES => 4-W PHOTOC. KL5 => … => 4-W PHOTOC. KL5 DOWN FULL/SW OFF (see user manual chapter 4.8.7)

7.23 Unauthoris. move

Error message:	(Status)* UNAUTHORIS. MOVE
Reason:	The door has performed an illegal movement although it should stand still.
Correctives:	1.)If this message is alternating displayed with the message „(Status)* POSITIONER“ the control system was not complete programmed. => Complete setup necessary. 2.)The brake is worn and slips („cannot hold“). Check function of brake and if applicable replace. 3.)Check brake connection and setting in the frequency converter (Brake selection power-on/power-off brake with sliding switch (P on / P off) and brake voltage with connection on KL2 (KL2; 1+3=200V, KL2; 2+3=100V).

	4.)Short circuit in 12 V control circuit (KL4, KL5;2 (Attention: KL5;2 test output 12V, KL5;1 24V)). Check cabling and connection for short circuit.

7.24 Door too fast

Error message:	(Status)* DOOR TOO FAST
Reason:	The door speed was min. 20% faster than the speed being learned.
Correctives:	1.) Check FC parameter in the control system. 2.) Adjust all end positions new. After this a new learning movement of the speed is automatically done.

7.25 Door to slow

Error message:	(Status)* DOOR TOO SLOW
Reason:	The door is significantly slower or not driven during the expected door movement.
Correctives:	1.) Check FC parameter in the control system. 2.) MCS-R: Check fuse F1-F3 in the control system and all three phases of the net supply. 3.) The brake does not release. MCS-FC: Check the brake voltage setting in the inverter at KL2. MCS-R: Check brake relay setting in the control system. Check brake rectifier in limit switch box of the motor.

7.26 Wrong direction

Error message:	(Status)* WRONG DIRECTION
Reason:	The door moves in the wrong direction.
Correctives:	1.) The temperature of the frequency converter is too high. The error message INV OVERHEAT is overwritten by the message WRONG DIRECTION. Reduce the number of door cycles. Mount frequency converter outside thermal interferences (sunlight, heat radiation drive unit). 2.) The brake resistor in the frequency converter is broken (200 Ω). To check turn of power supply. **Warning the frequency converter performs up to 180 seconds after turn off a dangerous voltage!** Open the frequency converter housing and disconnect both wires of the brake resistor from the board. Measure the resistance between both wires. If the value is significant greater or less than 200 Ω the resistor is broken and the complete frequency converter needs to be replaced. 3.) The door has abruptly stopped in the current movement, as when an error happens in the FC. The message is automatically reset after a short time.

7.27 Passdoor open

Error message:	(Status)* PASSDOOR OPEN
Reason:	Wrong parameter setting during the setup, because this input is not used on Albany ASSA ABLOY high speed doors. The pass door is open.
Correctives:	For disabling deactivate the input in the settings. SAFETY DEVICES => SAFETY 1 KL2 1-2 => … SAFETY DEVICES => SAFETY 2 KL2 3-4 => … Program according parameter list. (see also user manual chapter 4.8.4)

7.28 Reversed 3 times

Error message:	(Status)* REVERSED 3 TIMES This message will not appear in the display but only in the event log.
Reason:	The safety edge has been activated with the max. set numbers of activation in sequence. Thereafter the door is locked in the top position and there will be no more indication of the hold open time („----„).
Correctives:	1.) Remove objects from the doorline that the safety edge will not be activated. 2.) The bottom endposition was set too low. => Adjust bottom endposition. ADJUST ENDPOS. => ADJUST ENDPOS. BOTTOM => adjust. 3.) The safety edge was not disconnected in time at the bottom and the door reverse, because the safety edge is still active at the ground contact. Check the setting of the safety edge disconnection point and if necessary correct according to default parameters in the basic wiring diagram. ADJUST ENDPOS. => ADJUST ENDPOS. PRE END POSITION (see also user manual chapter 4.8.3) 4.) Broken wire or loose contact in the closing edge safety circuit during closing of the door. Check cabling of the safety circuit (prerunning photocell, safety edge).

7.29 Lift force det.

Error message:	(Status)* LIFT FORCE DET.
Reason:	Wrong parameter setting during the setup, because this input is not used on Albany ASSA ABLOY high speed doors. The set value of the force detection was exceeded.
Correctives:	To deactivate the function disable the lift force detection in the settings. MOTOR SETUP => MOTOR SETUP LIFT FORCE DET. => DEAKTIVATED. (see also user manual chapter 4.8.2)

7.30 Teach force det.

Error message:	(Status)* TEACH FORCE DET.
Reason:	Wrong parameter setting during the setup, because this input is not used on Albany ASSA ABLOY high speed doors. The learning movement of the slip detection must be performed manually.
Correctives:	To deactivate the function disable the learning movement in the settings. MOTOR SETUP => MOTOR SETUP LIFT FORCE DET. => DEAKTIVATED. (see also user manual chapter 4.8.2)

7.31 Error inv HW

Error message:	(Status)* ERROR INV HW
Reason:	Error in the frequency converter hardware or wrong motor connection.
Correctives:	1.) Turn off control system for min. 3 minutes (discharging time of the capacitor in frequency converter). After turning on power again the error has been reset automatically. 2.) Change parameter „MOTOR CONTROLLER“ first to „CONTACTOR/RELAY“ and then back to „INVERTER FS“. Turn off control system for min. 3 minutes (discharging time of the capacitor in frequency converter). After turning on power again the error has been reset automatically. 3.) Check motor connection in converter (KL1) and motor (X1). If short circuit has happened please eliminate first and then change converter. 4.) Check motor cabling and resistance of the motor winding. To check the motor windings remove plug X1 in the motor and measure on the counterpart of the plug in the motor (U,V,W and earth fault against PE). The 3 windings should all have the same resistance. If these clearly differ check the motor connection and if necessary replace drive unit.

7.32 Internal error

Error message:	(Status)* INTERNAL ERROR
Reason:	An internal error has occurred in the control system (unexpected event in the program / program execution error)
Correctives:	1.) Replace control system.

7.33 Motor unknown

Error message:	(Status)* MOTOR UNKNOWN
Reason:	The motor type is unknown, because it was not programmed during the setup.
Correctives:	1.) Set motor type in the control system. MOTOR SETUP => MOTOR SETUP FU PROFIL MOTOR => Select motor type according default parameter in the basic wiring diagram.

7.34 Not connected

Error message:	(Status)* NOT CONNECTED
Reason:	During the parametrization of the frequency convert the communication with the frequency converter was disturbed/interrupted.
Correctives:	1.) Check communication cable and plugs between control system (KL18) and frequency converter (KL4). 2.) Check voltage on control system output **(KL10; V, W)**. If there is no voltage, check net voltage on KL12; L1, N and internal fuses F1-F3. Replace broken fuses.

7.35 Waiting for FI

Error message:	(Status)* WATING FOR FI
Reason:	Info text that the control system waits after switching on for response from inverter.
Correctives:	1.) After all parameters are transmitted and the frequency converter is ready for operation the message is reset automatically.

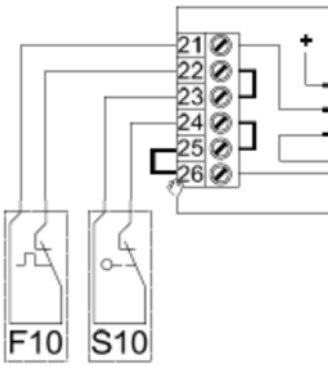
7.36 Inv overheat

Error message:	(Status)* INV OVERHEAT
Reason:	The frequency converter is overheated.
Correctives:	1.) Reduce number of door cycles. 2.) Mount frequency converter outside thermal interferences (solar radiation, heat radiation drive unit …).

7.37 Parameter upload failed

Error message:	PARAMETER UPLOAD FAILED
Reason:	The parameter for the frequency converter could not be successfully transmitted after 3 attempts from the control system to the converter.
Correctives:	1.) Check communication cable and connection between control system (KL18) and frequency converter (KL4). 2.) Confirm all parameters for the converter again complete and plug of net plug for min. 10 seconds.

7.38 Inverter error not connected

Error message:	INVERTER ERROR NOT CONNECTED
Reason:	The converter is not communicating with the control system or there is no voltage on the converter.
Correctives:	1.) Wire bridges on KL7 1/2 or KL20 4/3 are missing in the control system. 2.) The safety chain of the drive unit on KL8 was interrupted. The thermal contact from motor, handlever/crank and if applicable the hand chain drive are connected in the motor safety chain in serial (terminal 21-26 in the limit switch box of the motor). Unused terminals in the drive must be bridged.  F10= Thermal contact in motor S10= Emergency handlever/-crank Terminal 25, 26 not used and bridged. 3.) Check communication cable and plug between control system (KL18) and frequency converter (KL4). Pull out RJ-45 plug in the converter (KL4) and plug it in again. 4.) The supply to the converter was not connected to KL10 V, W (see basic wiring diagram).

7.39 Incompatible with inverter

Error message:	INCOMPATIBLE WITH INVERTER
Reason:	The selected digital limit switch (encoder) is not compatible with the frequency converter.
Correctives:	1.) Select the right digital limit switch. MOTOR SETUP => MOTOR SETUP LIMITSWITCH TYPE => Selection according default value in basic wiring diagram.

7.40 Overvoltage Inv.

Error message:	(Status)* OVERVOLTAGE INV.
Reason:	The DC-link voltage of the converter is too high.
Correctives:	1.) The brake resistor in the frequency converter is broken (200 Ω). To check turn of power supply. **Warning the frequency converter performs up to 180 seconds after turn off a dangerous voltage!** Open the frequency converter housing and disconnect both wires of the brake resistor from the board. Measure the resistance between both wires. If the value is significant greater or less than 200 Ω the resistor is broken and the complete frequency converter unit needs to be replaced.

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ASSA ABLOY Entrance Systems

info.aaes@assaabloy.com
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